

Sales & Profit Analysis Dashboard

using Power BI

Data Analytics & Business Intelligence Project
Project Report · 202

6.18M Total Sales	1.61M Total Profit	1.19K Total Orders	26.05% Profit Margin
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1. Introduction

This project focuses on analyzing retail sales data to understand business performance in terms of revenue, profit, customer behavior, and regional trends. The objective is to transform raw transactional data into meaningful, actionable insights using Microsoft Power BI — a leading business intelligence platform.

2. Objectives

- Analyze overall sales and profit performance across time periods
- Identify top-performing product categories and regions
- Detect loss-making sub-categories and recommend corrective measures
- Understand customer payment preferences and behavior
- Build an interactive, filterable dashboard for data-driven decision-making

3. Dataset Description

The dataset is a structured retail transaction log containing the following fields:

Field Group	Attributes
Order Details	Order ID, Order Date, Ship Date
Sales Metrics	Amount, Profit, Quantity
Product Details	Category, Sub-Category, Product Name
Customer Info	Customer ID, Customer Name, Segment
Payment Mode	Debit Card, Credit Card, UPI, COD, EMI

Location	Country, State, City, Postal Code
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4. Data Cleaning & Preparation

Prior to visualization, the raw dataset underwent a thorough cleaning and transformation process using Power Query:

- Identified and removed duplicate records to ensure data integrity
- Handled missing values using appropriate imputation or exclusion strategies
- Converted date columns (Order Date, Ship Date) to the correct DateTime format
- Created a calculated column: Profit Margin % = (Profit / Sales) × 100
- Validated and enforced correct data types for all columns
- Standardized categorical fields (State names, Category labels) for consistency

5. Tools & Technologies

Tool / Technology	Purpose
Microsoft Power BI	Primary BI tool for data modeling and visualization
Power Query (M Language)	Data ingestion, cleaning, and transformation
DAX (Data Analysis Expressions)	Custom measures, KPIs, and calculated columns
Microsoft Excel / CSV	Source data format and preliminary review

6. Dashboard Features

6.1 KPI Summary Cards

Four high-level KPI cards are displayed at the top of the dashboard, offering an immediate snapshot of business health: Total Sales (6.18M), Total Profit (1.61M), Total Orders (1.19K), and Profit Margin % (26.05%).

6.2 Visual Components

- Sales by Category – Horizontal bar chart comparing Office Supplies, Electronics, and Furniture
- Total Sales by State – Bar chart ranking the six states: New York, Florida, California, Texas, Illinois, Ohio
- Profit by Sub-Category – Detailed horizontal chart showing profit across 14 sub-categories
- Payment Mode Distribution – Donut chart segmenting Debit Card, Credit Card, UPI, COD, and EMI

- Year-wise Sales Trend – Line chart tracking annual sales performance from 2020 to 2025

6.3 Interactive Filters / Slicers

- Year filter (2020–2025) for time-period analysis
- State filter (California, Florida, Illinois, New York, Ohio, Texas)
- Category filter (Electronics, Furniture, Office Supplies)

7. Key Insights

The following actionable insights were derived from the dashboard analysis:

- Office Supplies leads in sales volume but Electronics shows the strongest profit margins, indicating a trade-off between volume and profitability.
- Certain sub-categories (e.g., Binders, Phones) show significantly lower profit relative to their sales, highlighting potential areas for price optimization.
- New York and Florida dominate total sales among all six states, suggesting higher market concentration in the northeast.
- Sales peaked around 2022–2023 and showed a notable decline in subsequent years, indicating possible market saturation or competitive pressure.
- Debit Card and Credit Card together account for over 40% of transactions, confirming the dominance of digital payment methods.
- Markers and Tables are the top two profit-generating sub-categories, contributing significantly to the overall 26.05% profit margin.

8. Conclusion

The Sales & Profit Analysis Dashboard built in Power BI successfully transforms raw retail transaction data into a comprehensive, interactive business intelligence tool. It empowers stakeholders to make data-driven decisions by revealing profitable product segments, underperforming regions, payment behavior patterns, and multi-year sales trends. The dashboard design prioritizes clarity and interactivity, enabling dynamic filtering by year, state, and category to support scenario-based analysis.

9. Future Enhancements

- Sales forecasting using Power BI's built-in analytics and Azure ML integration
- Customer segmentation analysis using RFM (Recency, Frequency, Monetary) modeling
- Automated data refresh pipeline via Power BI Gateway for real-time insights
- Mobile-optimized dashboard layout for on-the-go executive reporting
- Integration with CRM systems to correlate sales with customer acquisition data

End of Report